We look forward to seeing the results of applying these various methods in an actual clinical series. More importantly, as we indicate in our article, the ultimate goal is the development of accurate methods of measuring true corneal power.

DOUGLAS D. KOCH, MD
LI WANG, MD, PHD
Houston, Texas

Dear Editor:

I offer some insights regarding the data presented by Hamed et al[1] that should allow the application of this information for all cataract surgeons in a practical manner.

Although the clinical history method of determining the true central corneal refractive power was found to be the most accurate way of determining the true refractive power of the cornea,[1–3] this method requires that the keratometry values before the LASIK procedure be available. Since the time between the LASIK and cataract procedures will often extend to more than 10 years and the LASIK and the cataract surgery will often be performed by different surgeons, pre-LASIK keratometry will be difficult and sometimes impossible to obtain for many patients.

The contact lens method has also been proposed as an accurate way of determining the true corneal refractive power.[1–3] This technique is cumbersome to perform, which makes practical application of this technique in a busy cataract practice difficult. It is my experience that this technique is seldom used.

Computerized videokeratography can also be used to estimate the corneal refractive power; however, Hamed et al[1] have demonstrated that this will produce misleading values, as it uses a standardized corneal refractive index that is known to have been changed by the refractive surgery. Although adjustments in this value may increase the accuracy of this value,[4] the variety of corneal topography machines available will lead to confusion regarding the correct adjustment factors for each machine.

Based on the results of Hamed et al and my personal experience with this problem over the last 8 years, I offer the following recommendations for the adjustment of the keratometry value and hence the intraocular lens (IOL) power after LASIK:

1. The IOL power is calculated in the standard manner utilizing the keratometry values and the axial length.
2. The amount of LASIK refractive correction is obtained from the patient or medical records. This value is usually easier to obtain, as patients may be aware of their previous correction or have an old pair of glasses, or old LASIK records may be available.
3. Subtract 1 diopter from the mean measured readings for every 4 D of myopic LASIK correction. This is based on the assumption that the true central keratometry value will be approximately 75% less than the measured keratometry value. This percentage reflects the results of previous studies indicating that the 3.0-mm keratometry values underestimate the true refractive change at the cornea by 24%[5] and therefore indicate only 76% of the induced corneal flattening.
4. Aiming for a postoperative refractive goal of −0.50 to −1.0 D would introduce another margin of safety to avoid postoperative hyperopia but would also increase the LASIK enhancement rate. Postoperative myopia is a much preferred postoperative outcome as it will be better tolerated by the patients and, if necessary, much more easily corrected by a LASIK enhancement.
5. Patients should be counseled preoperatively that a small LASIK enhancement may be required after the cataract procedure.

Although these recommendations involve estimations, they should provide an easily applied practical method that all cataract surgeons can use with their existing equipment and formulas to achieve excellent postoperative cataract refractive results in most cases while avoiding the highly undesirable postoperative hyperopia surprise.

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Author reply

Dear Editor:

We thank Drs. Probst and Holladay for their comments. However, we are somewhat confused by their recommendations. In point 1, they recommend calculating intraocular lens power in the standard manner using the measured keratometry values. Then, in point 3, they recommend using readings that have been modified by subtracting 1.0 D for every 4.0 D of myopic LASIK correction. In fact, this latter recommendation is nearly identical to what we have recommended in our article, in which we suggest that readings be reduced by 0.24 D for every diopter of myopic LASIK correction. Therefore, for 4.0 D of LASIK correction, the keratometry reading should be reduced by 0.96 D.

Unfortunately, as we point out in our article, using keratometry readings is not as accurate as using the EffRP value that we determine from the EyeSys device (EyeSys Vision, Houston, Texas). Therefore, we hold to our recommendation to use the EffRP values whenever they are available. The keratometry values can be used if the EyeSys device is not available for measuring the patient.

The remainder of Probst and Holladay's recommendations are essentially identical to those that we made in our article.

DOUGLAS D. KOCH, MD
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Sensitivity and Specificity

Dear Editor:
The definitions of sensitivity and specificity provided by Choplin and Lundy[1] lead to nonsense. Consider this table:

	Disease		
Diagnosis	*Normal*	*Suspect*	*Glaucoma*
Normal	a	b	c
Suspect	d	e	f
Glaucoma	g	h	i

"Disease" indicates the true condition of the patient. Here, a through i are probabilities that sum to 1. Choplin and Lundy define false positive as (d+g+h), false negative as (b+c+f), sensitivity as (a+d+e+g+h+i), and specificity as (a+b+c+e+f+i). Let us assume that a = 0.8, b = c = 0.1, and all other cells = 0. According to Choplin and Lundy, the sensitivity would be 0.8, even though no subject with disease is ever detected. Now assume that a = 0, b = c = 0.4, d = g = 0.1, and all other cells = 0. According to Choplin and Lundy, the specificity would be 0.8, even though no normal subject is ever diagnosed correctly.

More reasonable definitions would be sensitivity = (e+h+i)/(b+c+e+f+h+i) and specificity = a/(a+d+g).

TERRY A. COX, MD, PHD
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Author reply

Dear Editor:
We appreciate Dr. Cox's interest in our work and the description of a tabulation for our data. It is particularly interesting to note that we had used this exact tabulation format in the original submission of our article! Perhaps we were not adept in describing the data in a clear and forthright manner, but the reviewers suggested a revision with alternative displays. The overall feeling was that the format using the tabulation suggested by Dr. Cox was too confusing.

We do not understand Dr. Cox's statement about no normal subject being diagnosed correctly using his model. If box a equals 0.8, then 80% of the time a normal patient was called normal (i.e., the specificity would be 80%), and 80% of the normals were therefore diagnosed correctly. Assuming boxes b, c, e, f, h, and i were indeed 0, there were no patients who were not normal and no statement could be made regarding sensitivity. In our population of normal subjects, those suspected to have glaucoma, and patients with glaucoma, this sort of distribution would not have occurred.

We do understand that our definitions of sensitivity and specificity are unconventional, and thank Dr. Cox for pointing this out. The false-positive rate should be how many times a normal patient is classified under "disease" (with specificity being 1 minus the false-positive rate) and the false-negative rate should be how often a patient with disease is called normal (with sensitivity being 1 minus the false-negative rate). These definitions are appropriate for studies in which cutoff values of single parameters are used in an attempt to discriminate between groups. Given that this was a clinical study in which masked observers were asked to make assessments of pooled printouts using no set criteria, we defined our rates with regard to the entire pool, and these definitions appear in the article. In other words, we defined false positives in terms of how many times a printout was incorrectly identified as abnormal and false-negatives as how many times a printout from an abnormal eye was called normal, within the context of the entire pool. Table 5 does use the more conventional definitions, with elimination of those suspected to have glaucoma to reduce diagnostic confusion. The demonstrated specificity was 79% with a sensitivity of 92%.

NEIL T. CHOPLIN, MD
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San Diego, California

Deep Sclerectomy in Refractory Congenital Glaucoma

Dear Editor:
In their retrospective series, Lüke et al[1] reported that deep sclerectomy was an ineffective procedure in refractory congenital glaucoma. Yet further examination of their data reveals that although they attempted "deep sclerectomy" in 10 eyes, in actual fact, they failed to do so in all but two patients. Out of 10 cases, 4 required conversion to trabeculectomy—one of whom had vitreous loss after iridectomy, 2 of whom required supplemental trabeculotomies, and 2 of whom had macroperforations during surgery. Of the remaining two eyes, "viscocanalostomy" as opposed to deep sclerectomy was performed, and these both developed hyphemas, which are supposedly uncommon with this procedure.[2] Thus, we can only examine these two eyes that apparently had "nonpenetrating" surgery to determine the effectiveness of such in this patient group. Surely this is too small a number to adequately answer the question of efficacy.

It appears that because "identification of Schlemm's canal was not feasible" and "percolation through the trabeculodescemetic membrane was too weak," nonpenetrating surgery was felt to be "not applicable" and required purposeful conversion to a penetrating technique in 6 of 10 cases. Yet, in another study of a similar technique in 12 eyes,[3] this procedure seemed applicable. In that study, Tixier et al found no intraoperative or immediately postoperative complications, and found the success rate to be equivalent to trabeculectomy, with diminished postoperative risks.

In congenital glaucoma, particularly in eyes with previous surgery, Schlemm's canal may be found as far as 5 mm posterior to the limbus and may be fairly scarred—further adding to the significant challenges in approaching these cases. In our experience, we have found it critical to initiate the deep scleral dissection more posteriorly than usual.

The authors state that the intraocular pressure–lowering